

Memory in Infancy and Childhood

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Abstract

Over the past five decades, there has been a tremendous effort to better understand the nature of memory in infancy and early childhood. These findings are beginning to address the puzzle of how it is possible to build an impressive semantic memory during periods of development when episodic memories for events are unlikely to be retained (infantile amnesia, 0-2 years) or are fragile and fragmentary (childhood amnesia, 2-8 years). We discuss three relevant issues: (i) how the nontrivial mnemonic capacities of infants fit with the facts of infantile amnesia; (ii) the contribution of changes in key component processes of episodic memory to the lifting of childhood amnesia between ages 2 and 8 years; and (iii) what we know about the development of the neural underpinnings of memory. We highlight the importance of the reciprocal relationship between research on memory development and theories and models of mature memory systems; that is, how current knowledge about development may constrain building viable models, and how contemporary models provide guidance for understanding memory development.

Keywords: memory development, memory, episodic memory.

Learning and memory affect all aspects of human development—perceptual, motor, cognitive, and social. Even before babies are born, they learn and retain information about the world, such as the sound profile of their mother’s voice. After birth, they acquire perceptual expectations such as the routines that guide visual scanning, and motor skills such as peddling a tricycle. They learn words like “cookie,” and they begin to remember (we hope) to say “please” when they ask for that cookie and “thank you” upon receipt of the cookie. Eventually, they recall specific events involving cookies, such as decorating cookies for a specific holiday at Granny’s house. These event memories become part of their personal identity and support planning for the future. Thus, understanding the development of memory is central to understanding human development. In addition, developmental data can and should contribute to the study of human memory in general, as they offer constraints on viable models of mature memory. However, the interdependence of research on children and the creation of memory models creates a classic “chicken and egg” problem. Cognitive researchers want to know how memory develops, but developmental researchers want a model of the mature memory system as a backdrop against which to ask questions about early capacities, as well as when and how they change.

This chapter cannot review the entirety of the past 50 years of research on the development of memory. We begin with a brief overview of two dominant memory models during (roughly) the period 1970-2010, and what lines of developmental memory research arose from them. At the end of this first section, we delineate the puzzle that is the focus of the rest of the chapter—namely, how it is possible to build an impressive semantic memory during periods when episodic memory for events is either lacking (infantile amnesia, 0-2 years) or fragile and fragmentary (childhood amnesia, 2-8 years). We discuss three pieces of this puzzle: how the non-trivial mnemonic capacities of infants fit with the facts of infantile amnesia; what the recent

literature on episodic memory development tells us about the lifting of childhood amnesia from 2 to 8 years; and what we now know about the development of the neural substrates of memory. At the end, we return to the theme of the inter-relationship between research on memory development and the construction of memory models, considering how current knowledge about development may constrain contemporary models of mature memory systems, and whether these contemporary models provide guidance for solving puzzles in memory development.

Memory Development and Memory Models: Two Waves of Research

When Flavell (1971) asked the question, “What is memory development the development of?”, the dominant framework for memory research was the Atkinson and Shiffrin (1968) model of information processing from sensory registers to short-term memory to long-term memory. Working within it, and using other ideas such as semantic integration (e.g., Bransford & Franks, 1971) and levels of processing (Craik & Lockhart, 1972), developmental psychologists gathered data on topics such as children’s iconic store and short-term memory (Morrison et al., 1974), and their ability to integrate information (Paris & Upton, 1976). They intensively studied the strategic and meta-mnemonic strategies that affect encoding into long-term memory and retrieval from it (Flavell, 1970), and children’s ability to monitor their strategies (Flavell, 1979). Research along these lines continues today, including studies of visual short-term store in infants (Oakes et al., 2006), working memory in infants and children (Cowan, 2016; Feigenson & Halberda, 2004; Gathercole, Pickering, Ambridge, & Wearing 2004), and the development of strategies for remembering facts and concepts, including how they affect educational outcomes (Fazio & Marsh, 2019; Ornstein, Coffman, Grammer, San Souci & McCall, 2010; Varga, Stewart & Bauer, 2016).

However, even in the 1970's, there were models of memory that differed from the Atkinson and Shiffrin (1968) framework. Tulving (1972) proposed a distinction between static knowledge of semantic facts and the temporal and spatial particularities of episodic memory, and Jacoby and Dallas (1981) contrasted episodic memory and perceptual learning. By the 1990's, most research on long-term memory began to use a multiple-memory-systems view (e.g., Squire, 1992), in which there is a distinction between declarative or explicit memory and procedural or implicit memory. In this way of thinking, Tulving's episodic and semantic systems nest within the explicit system, and Jacoby and Dallas' perceptual learning nests within the implicit system, along with classical conditioning, motor learning, and more. The source monitoring framework (SMF; Johnson, Hashtroudi & Lindsay, 1993), also formulated during the 1990's, augmented the multiple systems view by proposing that episodic memory involves binding various perceptual and conceptual elements into units (often events), with people making meta-mnemonic judgements at retrieval to decide whether an event really happened or was imaginary by evaluating the vividness and coherence of the memory. Adopting this emerging framework, developmentalists asked several new questions.

Implicit and Explicit Memory

When the idea of multiple memory systems appeared, there was considerable excitement among developmental researchers about the hypothesis that implicit memory develops earlier and/or lasts longer than explicit memory. There was a wave of developmental work on implicit processes as varied as classical conditioning (Stanton, Claflin & Herbert, 2010), motor learning (Koch et al., 2020); (Thomas & Nelson, 2001), perceptual priming (Drummey & Newcombe, 1995), and statistical learning (Saffran, Aslin & Newport, 1996). The data showed a wide variety of patterns of age-related change and long-term storage for these different kinds of implicit

memory (see review by Lloyd & Newcombe, 2009). Crucially, although there were instances of children showing preserved implicit memory with a lack of explicit memory (Perris, Myers & Clifton, 1990), it turned out that explicit recognition could also exist without implicit reactions (Newcombe & Fox, 1994), and that implicit memory declined over time, just like explicit memory (Lie & Newcombe, 1999). The bottom line seems to be that none of the various implicit memory systems constitutes a memory “permastore” that retains the traces of early experience in a robust fashion, as originally surmised.

Explicit Memory in Childhood

Research in the ‘70s and ‘80s produced a great deal of information about the development of school-age children’s ability to perform in traditional experimental paradigms such as free recall, including their development of strategic processes to improve memory. In the 1990s, propelled in part by translational efforts to evaluate children’s eyewitness testimony (Ceci & Bruck, 1993), there was a burgeoning effort to understand the development of memory outside as well as inside the laboratory, and an increasing focus on preschool-aged children from 3 to 6 years. Techniques used in these studies varied from the naturalistic (children’s memory for events of their lives; Fivush, Gray & Fromhoff, 1987) through the semi-naturalistic (memories for staged events; Principe & Ceci, 2002) to efforts to study event memory using more traditional methodologies derived from the SMF, such as source monitoring (Drummey & Newcombe, 2002) or reality monitoring (Lindsay, Johnson & Kwon, 1991; Sluzenski, Newcombe & Ottinger, 2004). This research showed that children as young as 3 years of age produce verifiable narratives of both naturally-occurring events and staged events, and start to deploy basic memory strategies (DeLoache, Cassidy & Brown, 1985; Balcomb & Gerken, 2008). Against this portrayal of early competence, there was also evidence that preschool children are more suggestible than

older children or adults (Bruck & Ceci, 1999), and are relatively poor at reality and source monitoring (Drumme & Newcombe, 2002; Lindsay et al., 1991; Sluzenski et al., 2004).

In short, although 3-year-olds demonstrate explicit memory, they are prone to systematic errors that render real-life uses of memory, such as eye-witness memory, problematic at this age. In addition, research shows that there is dramatic growth of explicit memory across preschool and into the early school years. Interestingly, research in this era did not clearly differentiate explicit memory into semantic and episodic memory. In retrospect, most of the paradigms were testing what we would call episodic memory, and thus these studies help delineate its developmental trajectory. Other paradigms, such as early narrative recall, produce unclear results. Preschoolers' memory reports are often so lacking in detail that they may either constitute an episodic but impoverished memory, or a schematic representation of a typical event, e.g. when asked about last week's zoo visit, a 3-year-old might respond "we saw animals". That is, these reports could be an example of either episodic or semantic memory.

Explicit Memory in Infancy

Research on memory during infancy proceeded in parallel with research on preschool children, but necessarily used very different techniques that did not rely on language. Two points of view emerged. One group of researchers made strong claims about the excellence of infant memory capabilities as evaluated by baby-friendly techniques, such as monitoring of eye movements (Diamond, 1990), deferred imitation (Bauer, 2004; Meltzoff, 1995) or operant conditioning (Rovee-Collier, 1997). In parallel, these researchers argued that the neural substrates of memory, in particular the hippocampus, are largely mature at birth (Diamond, 1990; Nelson, 1995). Seen through this lens, strong initial systems strengthen in continuously-developing processes, and language adds the potential for verbal recall, with no qualitative

differences in memory functioning (e.g., Bauer, 2006; Courage & Howe, 2004; Rovee-Collier, 1997).

An alternative view of infant memory questions whether infants can have true episodic memories. After all, older children and adults remember next to nothing from the first two years of their life, resulting in the phenomenon called infantile amnesia. A distribution of frequency of memories in relation to age at the time of experience, based on about 11,000 childhood memories gathered using various methods, shows almost no autobiographical memories predating the age of 2 years, and a very small number of memories between 2 and 3 years (Rubin, 2000; see Figure 1). Recent large-scale research has confirmed this pattern (Akhtar, Morrison, Justice & Conway, 2018). Although some respondents claim memories from infancy, they often produce generic or improbable accounts (for debate, see Akhtar, Justice, Morrison, Conway & Howe, 2019; Bauer, Baker-Ward, Krøjgaard, Peterson, & Wang, 2019).

The acquisition curve shown in Figure 1 contrasts starkly with the rapid acquisition of semantic knowledge during infancy. During the first two years of life, children learn many words (Bates et al., 1994; Frank, Braginsky, Yurovsky & Marchman, 2017; see Figure 2). They also become familiar with event scripts such as what happens at bedtime or on a trip to the zoo (Hudson & Nelson, 1986) and acquire gender labels and stereotypes (Weinraub et al., 1984). These disjointed developmental patterns suggest that acquiring explicit semantic knowledge does not depend on episodic memory, at least early in development. This hypothesis gains further support from data on children with hippocampal damage, who often appear cognitively normal until their teenage years due to their largely-intact ability to acquire semantic information, even though they show severe episodic memory impairments (Baddeley, Vargha-Khadem, & Mishkin, 2001; de Haan, Wyatt, Roth, Vargha-Khadem, Gadian & Mishkin, 2006; Gadian, Aicardi,

Watkins, Porter, Mishkin, & Vargha-Khadem, 2000; Vargha-Khadem, Gadian, & Mishkin, 2001; Vargha-Khadem, Gadian, Watkins, Connelly, Van Paesschen, & Mishkin, 1997). Adult caregivers, including teachers, may not become aware of these children's problems until their typically-developing peers start to remember past events with some richness and reliability.

Section Summary

The emergence of semantic memory before (or even without) episodic memory suggests the importance of delineating how the components of the episodic and semantic systems assemble over preschool and into the school years. This lead-lag relation also places constraints on models of the mature memory system (Keresztes, Ngo, Lindenberger, Werkle-Bergner & Newcombe, 2018; Newcombe, Lloyd & Balcomb, 2011). In the following section, we review what we know about the initial memory capacities of infants, arguing that, though impressive and explicit, they are not truly episodic. To preview, the signature elements of episodic memory seem to emerge between 18 and 24 months of age.

A Closer Look at Memory Development in Infancy

Some of the initial clues that memory in infancy might be largely a form of semantic memory came from research on spatial memory tasks that, like episodic memory tasks, also require the hippocampus; we will examine them first. These studies used paradigms originally devised for use in research on non-human animals, such as the Morris water maze. More recently, researchers have used nonverbal memory assessments—many also devised initially for non-human animals—to examine relational binding of what-where-when information with perceptual details to form an episodic memory. Both lines of work support the conclusion that *infant memory can be explicit, but it is non-episodic*. Of course, this idea sits uneasily with data attesting to the excellence of infant memory. In the second sub-section, we reconsider the classic

assessments of infant memory, arguing that performance initially lacks the full signature of episodic memory and that the key hallmarks of episodic memory capacities emerge in the second year of life. At the end of this section, we take a look at infant studies that use fMRI data obtained during sleep, which offer a means to supplement purely behavioral work.

Memory Paradigms Adapted from Research on Non-Human Animals

A rich source of techniques to use with nonverbal infants and toddlers comes from research with non-human animals, who also lack language. Of course, we must make many modifications to the paradigms used with animals, because we cannot deprive infants of food, raise them in controlled environments, or shock them. We also cannot give them the hundreds of training trials that non-human animals often require to learn a basic task. Nevertheless, there are many advantages to adapting the tasks used with non-human animals. They afford us the ability to utilize knowledge gained about neural bases of performance from animal experiments in which experimenters implant recording devices or lesioned specific areas, as well as to engage in cross-species comparison.

Spatial memory. Place learning is the ability to locate a position using information about its distance and direction from distal landmarks. One assessment is the Morris water maze, in which a rat is placed in a circular pool of murky water and must swim to a platform hidden just under the surface of the water. The hippocampal formation is required for success in the water maze when there are no landmarks marking the location directly, such as an object hanging directly above the platform and marking the spot (Morris 1983, 1984). The hippocampus also supports place learning in humans, based on studies of deficits in brain-injured human adults (e.g., Iaria, Petrides, Dagher, Pike & Bohbot, 2003; Maguire, Frackowiak, & Frith, 1997), and work using more direct recording techniques with epileptic patients (Ekstrom et al., 2003).

In children, place learning is first seen around 21 months of age (Balcomb, Newcombe & Ferrara, 2011; Newcombe, Huttenlocher, Drummey & Wiley, 1998). Earlier roots of place learning may be evident in simpler situations, in environments that do not require the use of distance and direction from several landmarks (Clearfield, 2004; Lew, Bremner & Lefkovich, 2000; Lew et al., 2004; Ribordy Lambert, Lavenex & Banta Lavenex, 2017). Place learning continues to develop well into the school years, with children being able to take advantage of more distal landmarks in more complex settings with increasing age (Laurance, Learmonth, Nadel & Jacobs, 2003; Leplow et al., 2003; Overman, Pate, Moore & Peuster, 1996). Nevertheless, the first emergence of recognizable place learning at 21 months is a strong clue that this age may be a time of important hippocampal maturation. Changes in performance on other spatial tasks known to have hippocampal bases also support a developmental shift toward increased hippocampal functioning at the end of the second year of life (Russell & Thompson, 2003; Sluzenski, Newcombe & Satlow, 2004).

Episodic-like memory. Work with non-human animals provides nonlinguistic methods to assess non-spatial hippocampal functions, including episodic memory. Tasks adapted from research with scrub jays (e.g., Clayton & Dickinson, 1998) are useful with children as young as 3 years (Hayne & Imuta, 2011; Russell et al., 2011) but are difficult to adapt for infants and toddlers. Another approach comes from work on contextually-contingent binding. For example, Eacott and Norman (2004) capitalized on rats' natural instinct to explore novel items by testing whether rats could detect changes in the overall configuration of objects in context, when the contexts and objects themselves were equally familiar. Because the rats had seen both objects in both locations, simple item recognition would not support noticing the novel placement of one of the objects, i.e., the what-where connection that is variable over context. Rats with lesions to the

fornix (the output pathway from the hippocampus) spent equal amounts of time exploring each object, whereas rats with intact fornices spent a significantly greater proportion of time exploring the novel object/context. Adapting work of this kind for use with human children, Newcombe, Balcomb, Ferrara, Hansen and Koski (2014) explored contextual episodic memory in toddlers and preschool children. Children explored two rooms, with the same four containers in each room but arranged in different configurations. Different people instructed them about the different rooms and what was hidden in the containers. One container in each room opened to reveal a distinctive toy, and the others were empty. For example, in the Rainbow Room, there might be a bottle of bubble blowing liquid in the black basket, but in the Cloud Room, there might be a ball in the red cube. After the exploration phase, we prompted children to search by handing them a very specific cue, namely part of the toy (e.g., the wand for blowing bubbles). Even with such a potent cue, as well as a short delay period, toddlers performed did not perform above chance until 20 months in this contextually-dependent task.

Traditional Infant Memory Assessments

Babies cannot speak or understand instructions to remember stimuli. Thus, the techniques that experimenters use to examine their memory are quite different from ones used with older children and adults. Infants actively look at stimuli they find salient or interesting, and can grasp and manipulate objects by around 6 months. In this section, we discuss three infant-friendly memory tasks based on these infant capacities, in which experimenters monitor infant eye movements, investigate whether babies remember situations in which their motor acts can create interesting effects, or check for deferred imitation of modeled sequences. For each task, we consider whether the task is explicit, and, if it does tap explicit memory, whether it is episodic. Evidence of an episodic-like memory can include retaining temporal order or spatial specificity,

showing relational or contextually-specific binding of elements, emerging after a single exposure, and showing retention over delay. Other kinds of evidence of hippocampal dependence can come from studies of adults with brain damage, studies of non-human animals, or studies of typical adults using measures of neural activation.

Eye Movements and Context. In a typical visual paired comparison (VPC) task, infants see pictures of a stimulus (often a face or object) until they start to look less at that stimulus (i.e., habituate), and they then are presented with the old stimulus and a novel one from the same category. The encoding trials can be quite extensive, and the retention interval is often quite brief. The experimenter measures how much infants look at novel as opposed to familiar stimuli; usually infants look longer at the novel stimulus, but after longer retention intervals, they may look longer at the familiar one, as if to reacquaint themselves with it (Bahrick & Pickens, 1995). Success in VPC is evident very early in life, but whether it shows explicit or implicit memory is unclear (see Rose, Feldman & Jankowski, 2007, for the case for explicit memory and Snyder, 2007, for doubts). Simple VPC also may not provide a sensitive behavioral index of episodic memory beyond the rudimentary. In an important variant of VPC, more likely to index hippocampal processing (Olivier Pascalis & Bachevalier, 1999; Pascalis, Hunkin, Holdstock, Isaac & Mayes, 2004), a picture of an object is presented on a solid-colored background, and the old and novel objects then appear with the background color constant or changed to another color. Babies do not succeed until 18 months of age in detecting the changed context (Robinson & Pascalis, 2004). Furthermore, research with infant monkeys with early hippocampal lesions and sham-operated controls shows a clear progression of performance on the VPC task extending to 18 months (of the monkey life span), with marked delay-dependent effects, especially pronounced in the lesioned monkeys at 18 months. Research suggests a developmental

progression in which initial recognition is of an object fused with a background (i.e., a unitized rather than a bound representation; Edgin, Spanò, Kawa & Nadel, 2014). Context effects initially disappear (in elementary school children) and reappear as object recognition strengthens and the objects are bound to separately-represented backgrounds. Lastly, although babies of 6 and 12 months look longer at stimuli shown against the original context than paired with new contexts (Chong, Richmond, Wong, Qiu & Rifkin-Graboi, 2015; Richmond & Nelson, 2009; Richmond & Power, 2014), these effects are fragile and fleeting, and appear to show a U-shaped developmental function, i.e., they do not appear in older children but reappear in adults, likely reflecting a different underlying dynamic (Koski, Olson & Newcombe, 2013; Edgin, Liu, Hughes, Spanò & Clark, 2019). In sum, research with VPC and related kinds of eye movement tasks does not support the idea that initial explicit memory is truly episodic. It progresses gradually to an adult state, with the initial inflection point in the second year of life.

Deferred Imitation. In a deferred imitation task, an experimenter displays a sequence of actions, and children can imitate if they wish after a delay. The length of the delay can vary, as can other variables such as how many actions are displayed, whether the order of actions is arbitrary or one depends on another, whether the child can also imitate immediately, and more. Adults with hippocampal damage do poorly on deferred imitation tests (McDonough, Mandler, McKee & Squire, 1995), as do people with developmental amnesia (Adlam, Vargha-Khadem, Mishkin & de Haan, 2005). Given this evidence, the developmental acquisition curve for successful performance on deferred imitation is a central issue in understanding the development of episodic memory in infancy. Fortunately, there is a well-documented sequence of development. Deferred imitation of a series of experimenter-modeled actions first appears at 6 months, albeit in simple forms involving short sequences, brief delays, or repeated retrievals at

the time of the initial encoding event (Barr, Dowden & Hayne, 1996; Barr, Rovee-Collier & Campanella, 2005). Thus, this ability is not definitively episodic nor does it necessarily involve the hippocampal system. Performance strengthens over the first 2 years of life, as children gain the ability to encode longer sequences, show longer retention, and require less specific cues for retrieval (e.g., Bauer, Hertsgaard & Dow, 1994; Bauer, Hertsgaard & Wewerka, 1995; Hayne, Boniface & Barr, 2000; Hayne, MacDonald & Barr, 1997). However, children often generalize their knowledge of sequences such as “how to build a rattle” over materials and situations, suggesting something more akin to semantic memory (Bauer & Dow, 1994; Lechuga, Marcos-Ruiz & Bauer, 2001). Crucially, it is only in the second year of life that children begin to exhibit the ability to encode and retain an arbitrary temporal sequence of actions, at 22 months on an immediate test (Bauer, Hertsgaard, Dropik & Daly, 1998) and at 28 months after a two-week delay (Bauer et al., 1998). Thus, as with VPC, the evidence from deferred imitation suggests an important inflection point towards the signature characteristics of episodic memory in the second year of life.

Conjugate Reinforcement. This operant task makes use of infants’ pleasure in controlling their environment and seeing interesting visual displays. Lying on their backs, infants look up at a distinctive mobile, which is attached to one of their ankles with a ribbon. Thus, the infant’s kicking causes the mobile to move. Rates of kicking when babies see mobiles that have formerly been controllable in this way exceed baseline kicking rates. Operant learning per se is not an index of episodic memory; the key element in this paradigm is that the unique perceptual characteristics of the mobile (and sometimes of the surrounding walls of the crib) influence the rates of kicking. Research has exhaustively tracked the ability of infants of various ages to retrieve a memory of the specific mobile they saw before, over various delays and with various

kinds of reminders, such as seeing the mobile move. Evaluated in this way, memory is present in the first few months of life, but becomes more robust over the following months (Rovee-Collier, 1997).

However, there has been some dispute over what kind of memory system is involved in this behavior. Some researchers have suggested it may be implicit (Bauer, DeBoer & Lukowski, 2007). Counter-arguments include the fact that these memories follow similar functional rules as explicit memories do in adults (Rovee-Collier, 1997); that performance tracks well with a more definitively explicit task that can be given by 6 months and continued in the second year of life, in which toddlers can activate a distinctive toy train by a method they need to retain over time (Hartshorn & Rovee-Collier, 1997); and that performance aligns well with deferred imitation (Gross, Hayne, Herbert & Sowerby, 2002). However, experimenters have not assessed contextual specificity by studying situations in which one action controls one stimulus and a different action controls a different stimulus. In addition, they have not assessed a situation in which infants must produce actions in an arbitrary temporal order. Lastly, we do not know the neural correlates of the task. Thus, despite the large literature on conjugate reinforcement, it is still unclear how to fit the data into an overall account of memory development.

Assessing Hippocampal Activation during Sleep

For all the measures discussed so far, we have only been able to assess their episodic nature using behavioral signatures, or data on analogous tasks with non-human animals, human patients, or typical human adults studied with fMRI. Prabhakar, Johnson, Nordahl and Ghatti (2018) offered a novel way to change that situation. They studied 2-year-old toddlers who had heard a novel song in a distinctive context and with a distinctive toy, during natural sleep in a scanner. They found that the toddlers showed greater hippocampal activation during the

previously-heard song than during a control song never heard before. This activation effect was especially marked for toddlers who were able to indicate the correct room and toy character associated with the song. Thus, it seems likely that the observed hippocampal activity involved reactivation of an episodic memory. Of course, these children were older than 2 years old, and were thus at the start of the period during which we believe that episodic memory formation is possible. It would be fascinating to study infants using this procedure. One extremely obvious fact about infancy and childhood is how rapidly the duration and organization of sleep change, especially during infancy (Huber & Born, 2014). The role of sleep in memory may change with age. Gómez and Edgin (2015) have argued that the evidence suggests that sleep promotes generalization of memories in infants but retention of specificity after the period of 18 to 24 months, in line with the general reorganization of the memory system at this time suggested by behavioral measures.

Section Summary

The available evidence converges on the conclusion that the distinctive elements of episodic memory emerge in the second year of human life. The memory abilities available earlier are certainly explicit but there is scant evidence they are episodic. They appear to be largely semantic, allowing children access to categories, concepts, and labels which can serve as the building blocks for later-emerging cognitive abilities, including episodic memory. In line with this hypothesis, the infant monkeys with hippocampal lesions in Bachevalier's experiments did not show initial memory problems relative to controls in infancy, but did so later in development. They "grew into" their deficits. However, we need more multi-method longitudinal work, more comparative work, and a finer componential analysis of memory (as discussed in the next

section). We may be able to gain a great deal of new information from emerging neural recording techniques suitable for use with infants and toddlers (Ellis & Turk-Browne, 2018).

Memory Development in Early Childhood

We have argued that there is an important inflection point for the growth of episodic memory between 18 and 24 months, probably not coincidentally associated with the offset of infantile amnesia. However, a milder kind of childhood amnesia continues through the preschool years and on to 8 years or so, as shown by the Rubin (2000) curve (see Figure 1). Furthermore, research on the binding of inter-item associations, such as unique pairings of an animal or an object with a background (e.g., dog-jungle, cat-desert), shows robust improvements during this period, which occur in the presence of less pronounced or nonexistent age-related differences in memory for single items (Lloyd, Doydum, & Newcombe, 2009; Sluzenski et al., 2006). Similarly, cross-sectional and longitudinal evidence on binding of facts with their source corroborates the idea that memory for pairings lags behind item memory from age 4 to 8 years (Drummey & Newcombe, 2002; Riggins, 2014).

In the past decade, this basic picture has been enriched by insights from newer computational models of memory that have begun to guide research on memory development, delineating a variety of specific processes that support distinct facets of episodic capacity. We begin this section with examining two basic aspects of episodic memory, *relational binding* and *temporal-spatial specificity*. We then progress to discussing the constructs of *pattern separation* and *pattern completion*. Based on studies of whether place cells remap as the shape of an arena slowly changes from square to circle (Lever, Wills, Cacucci, Burgess & O'Keefe, 2002), pattern separation and pattern completion are often considered opposite poles of a continuum. However,

conceptualization and assessment of the concepts in the area of memory research has been quite variable, and rather distinct from the spatial operationalizations, as we will discuss.

Relational Binding

Episodic memory involves more than pairwise relations, because particular memories are often contingent, e.g., as when we eat fish in a favorite restaurant when out to dinner with a friend, but eat pasta when at another dinner with a different friend. More recent studies have employed tasks that tap such complex, context-dependent, *relational* representations. We have already mentioned Newcombe et al.'s (2014) "two rooms" paradigm, which showed the dawning of relational memory around the end of the second year with a very potent reminder cue. Further work with the same paradigm involved giving children only contextual cues (e.g., the rainbow room, the person who had guided them in exploring the room, and a distinctive spatial arrangement of containers). In this situation, there was an extended developmental arc for selecting the contextually correct container, from very low performance around 2 years, to excellent performance by 6 years. Ngo, Newcombe and Olson (2018) and Ngo, Lin, Newcombe and Olson (2019) embedded a similar complex AB-AC relational structure (bear-book in the red house; bear-ball in the blue house) into a dynamic and engaging animation task that can easily be administered to much larger numbers of children than was possible with the two-rooms task. Findings showed that relational binding increased markedly from age 4 to 6 years, a mere two-year age difference.

This particular age range has been shown to be special in regards to relational binding by other investigators. Richmond and Pan (2013) studied 3- to 5-year-old children who learned two series of animal-place pairs (e.g. the duck likes the train station), each presented in a storybook. Every place was associated with two animals, each learned from one of the two storybooks. At

test, children were asked to choose the correct place paired with each animal in a two-alternative forced-choice test. Compared to 3-year-old children, 5-year-old children required fewer blocks to reach 70% accuracy, and were overall more accurate on this task. Yim, Dennis and Sloutsky (2013) asked 4-year-olds, 7-year-olds, and young adults to learn two lists of object pairs. Each list contained six pairs. Each pair consisted of an overlapping item (an object that occurred in both lists), and a unique item (an object that occurred only in one list). Participants learned all pairs in the first list until they reached 100% accuracy before learning the second list. Adults outperformed 7-year-old children, who outperformed 4-year-olds. Interestingly, increasing the salience of contextual cues improved 4-year-olds' performance, suggesting that poor attention to the context information at encoding may account for some of the age-related differences in learning relational information. Similarly, 4-year-olds can do quite well on relational binding tasks with low loads on working memory, but age-related differences appear with longer lists (Lloyd et al., 2009). The role of general cognitive factors in episodic memory development deserves more attention.

Temporal-Spatial Specificity

Some of the studies of relational binding involve the effects of spatial context, such as a different room determining which container to select to find a toy, or different houses contextualizing what a character likes to do. Other research has more directly probed the retention of spatial information, a crucial element in the *what-where-when* triad that characterizes episodic memory. Bauer, Doydum, Pathman, Larkina, Güler and Burch (2012) found steep change from 4 to 6 to 8 years in children's memory for what event occurred where in a series of staged events in an experimental room. Extending to look at temporal order information, and using a naturalistic set of parent-reported events, Pathman, Larkina, Burch and

Bauer (2013) found that 4-year-olds showed very poor temporal ordering, with some differences also present between 6- and 8-year-olds. Most recently, Mastrogiuseppe, Bertelsen, Bedeschi and Lee (2019) have extended the examination of temporal and spatial specificity to younger children. Their data suggest that what-where and what-when binding emerge first, at 3 years of age, and that they are a prerequisite to the capacity to form a three-way binding of what-where-when in episodic memory.

Pattern Completion

A defining feature of episodic memory is that complex and multi-modal events are stored as coherent representations, so that episodic retrieval entails the holistic re-experience of all constituents of an event (Tulving, 2002). This coherence allows for the recovery of the entire event based on a partial cue so that all elements within an event can be retrieved successfully (or not), a process termed pattern completion. One approach to studying pattern completion is to examine retrieval dependency among different associations within the same event. The idea of retrieval dependency, defined as the contingency of retrieval success among within-event associations, was demonstrated in young adults by Horner and colleagues (Horner & Burgess, 2013, 2014; Horner, Bisby, Bush, Lin & Burgess 2015; see also, Chapter 5.7). In their paradigm, participants learn a series of unique events, each comprised of a scene, a person, and an object. Subsequently, participants perform a cue recognition task on every possible cue-test pair of each event. Successful retrieval of one association (e.g., retrieving a person when cued with a scene) was statistically related to the retrieval success of other associations from the same event (e.g., retrieving an object when cued with a scene), providing evidence for the idea of holistic episodic recollection. Although conceptually related to relational binding, holistic retrieval via pattern completion adds the element of contingency of retrieval success, based on encoding events as an

interwoven network of elements. Pattern completion may be basic to recollection, but it is a process that can be applied to various kinds of content-based representations—in this case, complex, high-dimensional, associative ones (Cowell, Barense & Sadil, 2019).

Very little is known about the development of holistic recollection. In an initial study, Ngo, Horner, Newcombe and Olson (2019) assessed coherence of within-event retrieval in 4-year-olds, 6-year-olds, and young adults by adapting the multi-element event task to a child-friendly format. Interestingly, retrieval dependency was evident as early as age 4, suggesting that even young children are capable of retrieving multi-element events as integrated units. However, the degree of retrieval dependency increased from age 4 to young adulthood. Dependency was greater in adults than in 4-year-old children, whereas dependency in the 6-year-olds was intermediate—it did not differ significantly from either the 4-year-olds or young adults. These results suggest that pattern completion is present in early childhood and undergoes critical refinements between early childhood and young adulthood.

Pattern Separation

Past experiences can share a great deal of overlapping content, hampering the retrieval of a specific event against the backdrop of interfering memories. Thus, successful memory encoding would be optimized by representing similar experiences using non-overlapping neural populations in the service of preserving each event's distinctiveness. Pattern separation aids mnemonic discrimination by reducing the degree of overlap between similar inputs (O'Reilly & McClelland, 1994; Norman & O'Reilly, 2003). One paradigm posited to estimate the behavioral expression of pattern separation, but that also places demands on pattern completion processes, is the Mnemonic Similarity Task (MST; reviewed in Yassa & Stark, 2011). In this task, participants first view a series of object images at encoding. Subsequently, the test list includes

targets (identical to the objects seen at encoding), lures (similar exemplars of the objects seen at encoding) and foils (new and dissimilar objects from those seen at encoding). The key aspect of this task is that it requires the formation, storage, and retrieval of high-resolution representations of studied objects in order to correctly discriminate similar lures from studied items.

At age 4, children are more likely to misidentify lures as studied items, which reflects a failure, in part, of pattern separation (Ngo, Newcombe, & Olson, 2018; Ngo et al., 2019; see Figure 3). This interpretation is based on the fact that these children called lures “old” more often than “similar” or “new”, suggesting that they retrieved the learned items, but with low precision. The ability to form and retrieve high-resolution memories continues to improve up to age 6 and beyond (Canada, Ngo, Newcombe, Geng, & Riggins, 2018; Ngo et al., 2018; Keresztes et al., 2017; Rollins & Cloude, 2018). Importantly, there is evidence that even the youngest children comprehend the task demands of the MST, based on pretest practice sessions (Canada et al., 2018; Ngo et al., 2018; Rollins & Cloude, 2018) and the fact that even 4-year-olds showed a correlation between lure discrimination and level of similarity (Ngo et al., 2018). Augmenting children’s motivation to encode and retrieve the specific details of the studied objects using gain-loss framing boosted preschoolers’ mnemonic discrimination, but did not abolish the robust age-related differences between preschoolers and their older counterparts (Ngo, Newcombe & Olson, 2019). The persistent age-related improvements argue for underlying change in the core ability of episodic memory, perhaps due to neurobiological maturation of regions that instantiate pattern separation and pattern completion.

Pattern separation should apply to contexts as well as individual objects. Surprisingly, context discrimination has been even less studied than item discrimination in developmental populations, but is imperative given the pivotal role of spatial context in episodic memory

(Robin, 2018). We recently examined this issue by varying the similarity of encoding contexts in which AB and AC associations were learned to probe relational binding with differential context pattern separation demands (Ngo et al., 2019). Four-year-olds and 6-year-olds performed similarly on the relational binding task (but worse than young adults) when spatial contexts were highly similar, i.e., when pattern separation demands were present for contexts rather than for local objects. While noting that paradigms designed to vary pattern separation demands likely also place variable demands on pattern completion, collectively these findings suggest that the maturation of pattern separation for spatial context is more prolonged than for individual objects.

One Line of Development, or Independent Lines?

Relational binding, temporal-spatial specificity, mnemonic discrimination and holistic recollection show strikingly similar patterns of age-related growth, with marked improvement between 4 and 6 years, and continued change to 8 years in some more complex dimensions, such as the discrimination of similar contexts. These data support the idea that childhood amnesia lifts only gradually and augment our understanding of how change occurs, based on componential analysis of the processes involved. Surprisingly, despite similar profiles of age-related change, there is apparently not one underlying dimension of growth for all the various lines of development. Performance on relational binding and mnemonic discrimination do not correlate, in several samples using varying techniques (Keresztes et al., 2017; Ngo et al., 2018; Ngo et al., 2019). This null correlation holds even when the two tasks share the same encoding phase, and are tested in the same 4AFC test format (Ngo et al., 2019). The weak correlation between relational binding and mnemonic discrimination in children may reflect, in part, their differential interaction with other cognitive systems relevant for the tasks, such as attention and strategic processes. However, it is important to note that in the same study, individual differences in

relational binding and mnemonic discrimination strongly correlate, suggesting that the uncorrelated performances in children and adults are not driven by process-irrelevant psychometric properties of the tasks.

Further, when Ngo et al. (2019) manipulated levels of task difficulty to yield significantly different pairwise associative memory performance in 6-year-olds, the same level of retrieval dependency was observed, revealing an intriguing dissociation between pairwise relational memory and holistic recollection. Lack of correlation suggests parallel development of independent processes across this age range, i.e., different children develop distinct facets of episodic memory capacities in the same age window, but at varying rates for each child. Not all the component processes have been examined together, but the emerging picture is one of rapid change along parallel lines.

Section Summary

Research in the past decade has been increasingly componential, and more intertwined with advances in our understanding of the neural substrates of memory and changes in models of mature memory, as we discuss in the next sections of the chapter. A schematic depiction of successive waves of developmental plasticity is shown in Figure 4, taken from Alberini and Travaglia (2017). There is still a great deal to learn about the descriptive behavioral facts and to bring findings from rodent and human data into closer alignment. Crucially, recent research has begun to suggest that the multiple converging lines of change are perhaps proceeding independently. If verified, we will need also to understand how they come to work together. Furthermore, we will want to find out about individual differences in rates and profiles of progress, and their biological and environmental correlates and consequences.

Development of the Neural Substrates of Memory

Behaviorally, as we have seen, episodic memory emerges in the second year of life and undergoes subsequent robust improvements to various components. Not surprisingly, given the behavioral findings, accumulating data have challenged the conclusion of earlier reviews that the hippocampus is mature in infancy (Diamond, 1990; Nelson, 1995). In fact, the neural substrates of the memory system undergo protracted developmental change, as reflected in more recent reviews (Richmond & Nelson, 2007; Riggins, Canada & Botdorf, 2020). As with behavior, change to the hippocampus occurs along multiple dimensions. Research initially focused on measures of overall hippocampal volume, but we are now beginning to learn more about changes in the volume of hippocampal subfields, and about rates of neurogenesis and its effects. Hippocampal connectivity is also important, both in terms of the internal connections between subfields that allow the hippocampus to act as a coherent unit, as well as the maturation of fiber pathways that provide the hippocampus with highly-processed sensory input, and neuromodulatory pathways that control the excitation and inhibition of this structure. In this section, we examine the developmental curves for these neural indices, as well as what we know about the specific behavioral correlates of each kind of change.

Hippocampus

Hippocampal volume increases dramatically in the first two years of life, aligning with the offset of infantile amnesia (reviewed in Newcombe & Olson, 2014). Growth then levels off, although some studies show that volume increases at a slow rate through childhood and perhaps into early adolescence (Østby et al., 2009; Utsunomiya, Takano, Okazaki, & Mitsudome, 1999; Canada, Botdorf, & Riggins, 2020; see Figure 5), while others do not (Giedd et al., 1996) or at least not in adolescence (Yurgelun-Todd, Killgore, & Cintron, 2003). Behavioral correlates of hippocampal volume are not clear; one study reported that episodic memory correlated with

hippocampal volume in 6-year-olds but not in 4-year-olds (Riggins, Blankenship, Mulligan, Rice, & Redcay, 2015).

Hippocampal subfields are increasingly important in current thinking about memory. Research on hippocampal subfields in monkeys indicates that the monosynaptic pathway (monosynaptic projections from entorhinal cortex [EC] to CA1 and back) develops earlier than the trisynaptic pathway (three synapses: from EC to dentate gyrus [DG] to CA3, and then to CA1; Gómez & Edgin, 2016; Lavenex & Banta Lavenex, 2013). There may be heterogeneous maturational rates among different hippocampal subfields in humans too (Daugherty, Yu, Flinn, & Ofen, 2015; DeMaster, Coughlin, & Ghetti, 2016; Gogtay et al., 2006; Keresztes et al., 2017; Lee, Ekstrom, & Ghetti, 2014). Gray matter volume of the late-developing subfields, including the DG and CA3, appears to follow a quadratic relation with age. The granule cells in the DG are thought to assign distinct representations to similar inputs via sparse coding, thereby amplifying small differences from the entorhinal cortical input to minimize interference in hippocampal memory representations (McClelland et al., 1995). In other words, pattern separation aids mnemonic discrimination by using distinctive neural codes to represent similar memories. Lure discrimination appears to relate to the DG/CA3 signals of the hippocampus, such that large responses are evoked by small changes in perceptual input (Lacy, Yassa, Stark, Muftuler, & Stark, 2010; Reagh & Yassa, 2014; reviewed in Yassa & Stark, 2011). Behaviorally, gray matter volume of the late-developing subfields (DG/CA2-CA4) relates to improvements in mnemonic discrimination performance from ages 4 to 8 (Canada et al., 2018). Holistic recollection may be supported by recurrent collaterals in CA3, which seem to form an auto-associative network that permits disparate inputs to be bound together, allowing for the recovery of the entire event based on a partial cue (Grande et al., 2019). Further, one study that used multivariate analysis

techniques to express the interrelated pattern of age-related differences in hippocampal subfields showed that the uneven maturational trajectories of hippocampal subfields are related to an improvement in mnemonic discrimination from ages 6 to 14 (Keresztes et al., 2017).

Neurogenesis is another important aspect of neural development, especially in the dentate gyrus. Ramsaran, Schlichting and Frankland (2019) summarize evidence that changes in rates of DG neurogenesis may underlie the offset of infantile amnesia, based on data from rodents and monkeys, who also exhibit infantile amnesia. During the first approximately two years of human life, new neurons are added to the dentate gyrus at rates about 100 times greater than in the adult dentate gyrus. These new neurons could destabilize memory traces, making it impossible to form long lasting memories. When hippocampal neurogenesis declines after age 2, the ability to form more long lasting memories strengthens. Most studies in non-human animals have used fear conditioning, which is unfortunate, because we cannot obtain comparable data from human babies. However, Travaglia, Steinmetz, Miranda & Alberini (2018) have recently confirmed important aspects of the findings, using an object location task with rat pups at 17 postnatal days. Interestingly, 17 days is in the middle of the organization of the navigation system in the developing rat, after neurons that increase firing rates for specific head positions, i.e. head direction cells, and neurons that increase firing rates for specific locations, i.e. place cells, begin to function, but before a more complex type of place cell called a grid cell (Tan et al., 2017) begins to function. This timing likely corresponds roughly to the middle of the second year of human life. Thus, it is possible that the timelines for the neural substrates of episodic memory and navigation coincide closely, as one might expect.

Beyond the Hippocampus

The *parahippocampal* and *perirhinal* cortices traditionally combine with the hippocampus to form the medial temporal lobe memory system. In addition, *retrosplenial* and *entorhinal* cortical areas are important in many contemporary memory models, which go considerably beyond the claim that the hippocampus is centrally responsible for episodic memory (Davachi, 2006; Diana, Yonelinas & Ranganath, 2007; Sugar & Moser, 2019; Ranganath, 2010; Robin, 2018). Ranganath and Ritchey (2012) suggest that there are two broad memory systems: (i) the anterior temporal system, which processes objects and people and can be thought of as a “semantic” system, and (ii) the posterior medial system, which processes the relationships between entities, actions, and outcomes, otherwise defined as an “episodic” system. This idea is interesting from a developmental standpoint given that semantic memory develops much earlier than episodic-like memory. Unfortunately, however, we know very little about the development of these other medial temporal lobe (MTL) areas. There is some evidence that parahippocampal cortex undergoes extended functional development even after the age of 7 years (Golarai et al., 2007; see also Ofen, Chai, Schuil, Whitfield-Gabrieli & Gabrieli, 2012). Given the importance of scenes and spatial context for episodic memory (Robin, 2018), we clearly need to know much more about very early development of the posterior medial system, which includes parahippocampal, retrosplenial and entorhinal cortices (Ranganath and Ritchey, 2012).

Maturation of fiber pathways that link the hippocampus to other structures must also be important in episodic memory development. There are many tracts to consider, most notably the tracts connecting the extended hippocampal system to the prefrontal cortex (PFC). (Of course, frontal cortex itself changes throughout childhood, which is also likely vital for memory functioning, especially in terms of the use of strategies and meta-memory; Bauer, 2007;

Tsujimoto, 2008; Yeatman, Wandell, & Mezer, 2014). Functional connectivity between the MTL and PFC increases through middle childhood (Menon, Boyett-Anderson, & Reiss, 2005; Ofen, et al., 2012), although we know little about earlier periods. We know even less about structural connectivity of the MTL. Extrahippocampal regions may play a critical role in transmitting hippocampal signals to the frontal lobe. The *uncinate fasciculus* connects the temporal pole, perirhinal cortex, and the amygdala to lateral orbitofrontal cortex, and would constitute part of the extended anterior temporal memory system as proposed by Ranganath and Ritchey (2012). This large fiber pathway is important for some types of semantic memory functions, such as learning detailed semantic associations (e.g. proper name associations, word pair associations) and imbuing concepts with emotional tone (reviewed by Von Der Heide, Skipper, Klobusicky, & Olson, 2013; see also Metoki, Alm, Wang, Ngo, & Olson, 2017). This tract has a protracted developmental profile, continuing to develop through adolescence and even into early adulthood (Von Der Heide et al., 2013). The *fornix*, a C-shaped white matter tract that creates a monosynaptic pathway between the hippocampus and several subcortical regions (Amaral & Lavenex, 2007), is also important. Damage to this tract causes severe and unremitting anterograde amnesia that bears a striking similarity to amnesia caused by hippocampal pathology. The fornix can be identified in the fetal brain within the second trimester, and the entire fornical topology is developed at 19 gestational weeks (Huang, 2010). However, findings are mixed regarding microstructural changes in axonal thickness, density, and organization in early childhood (for a review, see Benear, Ngo, & Olson, 2020), with some studies showing developmental changes (Reynolds, Grohs, Dewey, & Lebel, 2019; Canada, Botdorf, & Riggins, 2020) and others not (Lebel, Walker, Leemans, Phillips, & Beaulieu, 2008; Ngo et al., 2017).

Section Summary

Far from being mature in infancy, memory-relevant neural areas and pathways undergo considerable change across development, extending well into early childhood. Investigation of many areas and pathways is very sparse, however, along with exactly how each behavioral component relates to each neural component. We can expect to see an explosion of knowledge in the field as more refined behavioral assessments are paired in longitudinal studies with better imaging and data-analytic techniques, as enabled by the recent consensus on hippocampal subfield delineation (Wisse et al., 2017).

Memory Development and Memory Models: Current Status

We opened this chapter by sketching out the idea that semantic memory develops considerably in advance of episodic memory. Subsequently, we have focused mainly on delineating the protracted and componential development of episodic memory. Returning to the puzzle with which we started, and the two-way street connecting modeling to developmental data, in this section we consider two topics. First, we look at the common assumption in many memory models that semantic memory arises from abstraction over multiple episodic memories, and the challenges to such models coming from developmental findings. Second, we consider what contemporary models, which have moved beyond the simple multiple-systems models of the 1990s, suggest about how babies and toddlers may build semantic memory while retaining only fragile and fragmentary memories of individual episodes.

Challenges to Memory Models from Developmental Data

In classic consolidation theory, memories are initially stored in the hippocampus, but over time are offloaded and stored only in the neocortex (e.g., Alvarez & Squire, 1994). The complementary learning systems (CLS) approach proposed by McClelland, McNaughton & O'Reilly (1995) provided a useful computational framework for thinking about the utility and

functioning of this architecture, highlighting a fundamental tradeoff between encoding specific details tied to individual experience, and extracting commonalities across experiences. The CLS posited that the hippocampus performs computations that support pattern-separated and individualized memories, and then slowly transfers these experiences to cortical areas during “offline” periods such as sleep. In contrast, the cortex (the particular part of the cortex was never defined) has a slow learning rate (also not defined) and allows commonalities in overlapping representations to be extracted, leading to a more general “semanticized” memory. Some variant of the assumption of abstraction is still common today (e.g., Klinzing, Niethard & Born, 2019). Indeed, it is also present in multiple trace theory (MTT), the most prominent alternative to standard consolidation theory (Renoult, Irish, Moscovitch & Rugg, 2019). On any of these accounts, how do we explain the absence of event memories from the first two years of life? How is it possible to build semantic memory in a developmental window when long-lasting episodic memories are less likely to be retained?

One possible model more consistent with the developmental facts is to suggest that semantic memory and episodic memory rely on distinct computations in distinct brain regions—semantic memory depends on a network of distributed sensory-specific brain regions and left-lateralized “hubs” in the anterior temporal lobe and angular gyrus (Binder & Desai, 2011), while episodic memory depends on the extended hippocampal system (Mishkin, Suzuki, Gadian & Vargha-Khadem, 1997). Although Renoult et al. (2019) point out that the general semantic network and the episodic core recollection network share a substantial amount of neural territory, other data suggest that these systems may be dissociable (Hodgetts et al., 2017). This view, which we elaborate on in the next section, can accommodate developmental findings about the

uneven development of episodic and semantic memory in childhood better than classic consolidation theory or multiple trace theory.

Clues to Development from Memory Models

We propose that there are two neural pathways for acquiring semantic memories: a purely cortical route that is relatively slow and relies on repetition, and a fast (in some cases, one-trial) hippocampal route. It is important to note that the acquisition of vocabulary by infants and toddlers is typically quite slow, contrary to a widespread assumption that it occurs via a “fast mapping” process (O’Connor & Riggs, 2019). The vocabulary spurt that takes children rapidly past the first 50 or 100 words occurs at 18 to 24 months, coming at roughly the same time as other indices of hippocampal function. In this view, as the hippocampus starts to come online, semantic learning becomes more rapid, and less reliant on repetition. Likewise, adults with anterograde amnesia show semantic learning that is slow, requires repetition, and is reliant on preexisting semantic knowledge (Hebscher, Wing, Ryan & Gilboa, 2019). In contrast, such learning is rapid in neurologically-intact adults, who already have a rich semantic store that can serve as an associative scaffold for learning new-but-related concepts.

The puzzle for development is how such learning works with a very lean stock of preexisting knowledge. One possibility is that proto-episodic memory is capable of retaining some information over short delays and when combined with environmental support—for example, through densely-spaced teaching events or synchrony of infant attention with caregiver input, children can build general concepts within short time windows (Clerkin, Hart, Rehg, Yu & Smith, 2017; Stahl & Feigenson, 2017). Another possibility involving the environment is that the largely familiar environments in which children spend the bulk of their time in the first years

of life differentially support the retrieval of semanticized and general memories, thereby strengthening them, as found in adults by Robin, Garzon and Moscovitch (2019).

Another interesting possibility is that different hippocampal circuits carry out different computations. Schapiro, Turk-Browne, Botvinick and Norman (2017) speculate that the monosynaptic pathway described earlier supports statistical learning, while the trisynaptic pathway supports pattern separated representation of similar experiences. This view is speculative, given the uncertain role of the hippocampus in statistical learning (Jiang, Sisk, & Toh 2019), and also in light of the fact that we know next to nothing about these small but important fiber pathways in humans. However, this view does raise the interesting possibility, so far untested, that rapidly-acquired semantic memory might also rely on an early-maturing monosynaptic pathway in the hippocampus, in addition to a cortical system. In fact, Sucevic and Schapiro (2019) conducted simulations that showed that a model of learning in hippocampal subfield CA1 allows for learning of categories and generalizing to novel instances. However, future research needs to evaluate whether there is a behavioral reality to this in-principle possibility that the monosynaptic pathway, with its relatively slow learning rate and overlapping representations, can extract regularities underlying category structure.

Looking Ahead

Although we know an impressive amount about memory development, many empirical and conceptual challenges remain. Empirically, there is a fairly long to-do list. We need much more methodological work on assessments: what capacities infant measures actually tap (usefully informed by continued development of common paradigms for comparative work with non-human animals); relations among various facets of episodic memory during childhood; and generalization of experimental measures to real-world autobiographical recall (for one start, see

Pathman, Samson, Dugas, Cabeza & Bauer, 2013). We need much more knowledge about the neural correlates of memory development, including MRI work investigating sleep or even wakefulness in babies, relations of behavior to high quality structural measures in children, and the nature of functional reactivation across infancy and childhood. Furthermore, the field would benefit from comparisons of generalized knowledge and episodic memory using similar acquisition paradigms (e.g., Ngo, Benear, Popal, Olson, & Newcombe, 2020, preprint). Finally, there are new areas of investigation to open, such as the role of curiosity and motivation in memory development (Gruber & Ranganath, 2019; Murty & Dickerson, 2016), how the components of memory are affected by environmental adversity, and how memory development relates to academic achievement. Conceptually, developmentalists and memory modelers both need to tackle the precise nature of the episodic-semantic distinction. Renoult et al. (2019) address this issue, but without considering what the semantic-before-episodic nature of development tells us about the nature of the distinction. Future work will benefit from closer communication between separated intellectual communities.

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Figure 1. The distribution of memories as a function of age. This graph demonstrates the rapid increase in autobiographical memories during very early childhood and reaching maturation around 7 years of age. Adapted from Rubin, 2000.

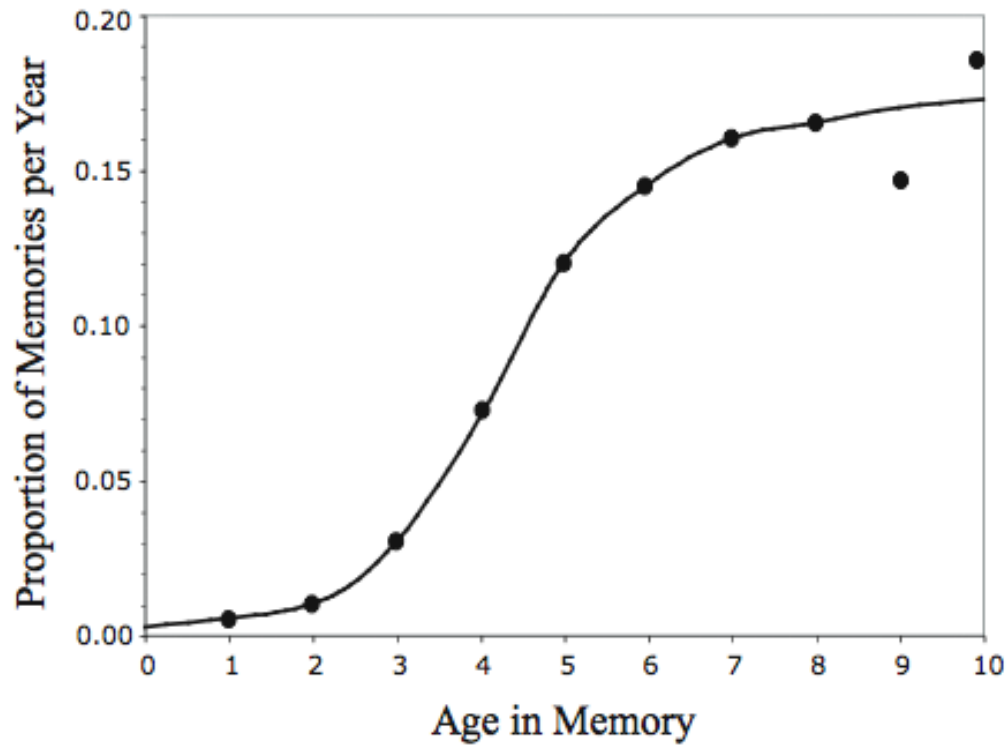


Figure 2. Children's productive vocabulary growth curve. Data on children's vocabulary acquisition, depicted with using The MacArthur-Bates Communicative Development Inventories (CDIs) Vocabulary Norms analysis tool. The graph shows the 10th, 25th, 50th, 75th, and 90th percentiles for English production scores. Dots represent individual children, jittered slightly to avoid overplotting. Curves show polynomial spline fits. Adapted from Frank, Braginsky, Yurovsky & Marchman, 2017.

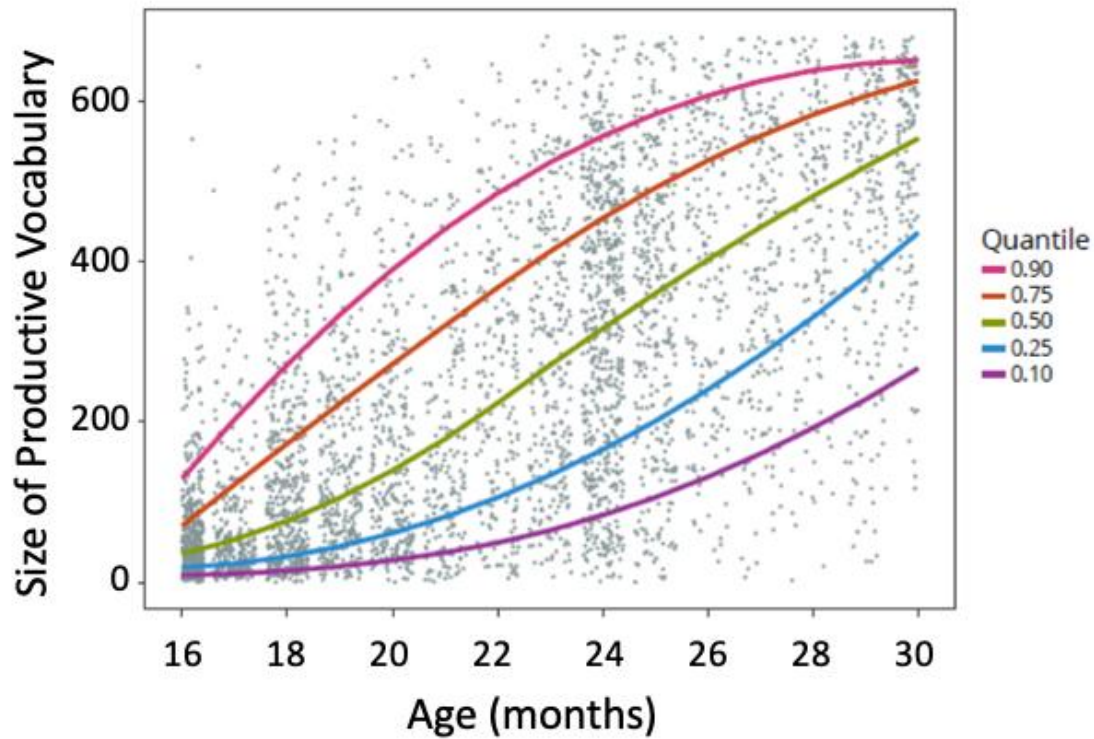


Figure 3. A schematic depiction of the encoding and test phases of the child-friendly Mnemonic Similarity Task (Ngo et al., 2018; Canada et al., 2018). LEFT: is a depiction of the stimuli and the task. Participants were told that they would play a game about pictures of objects. The task was to look at pictures of objects presented one at a time on a computer screen and to make an indoor/outdoor judgment for each object. At test, participants were asked to decide whether the presented item was identical, similar, or new, compared to what they had seen previously. MIDDLE: data from Ngo et al., (2018) illustrating robust age-related improvements in pattern separation performance, indexed by lure discrimination ($p[\text{similar}|\text{target}] - p[\text{old}|\text{target}]$); RIGHT: data from Canada et al. (2018) using the same task and a larger range of ages, ($p[\text{similar}|\text{lure}] - p[\text{similar}|\text{foil}]$).

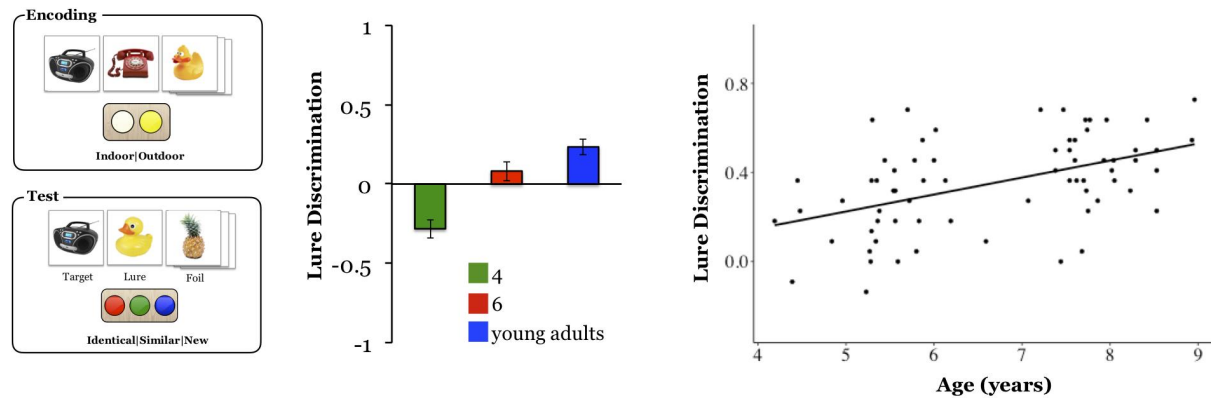


Figure 4. A schematic depiction of the development of memory, from Alberini & Travaglia (2017).

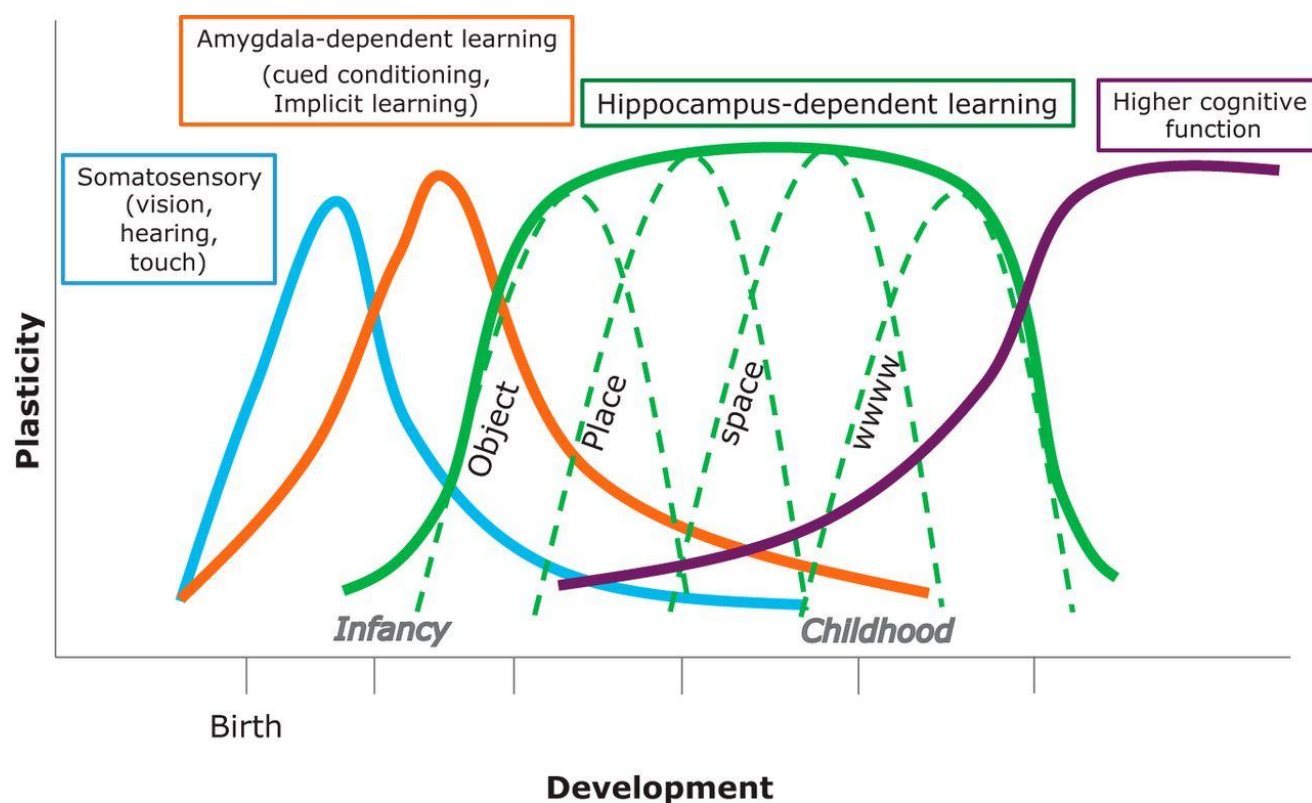


Figure 5. Developmental changes in the ICV-adjusted volume of hippocampus. Graphs depict bilateral (top panel), left (middle panel), and right (bottom panel) (A) total hippocampus; (B) hippocampal head; (C) hippocampal body; and (D) hippocampal tail. Error bands represent 95% confidence intervals. * denotes significant effect of age at $p < .05$ level, and ~ denotes marginal effect of age at $p < .06$ level. Age^3 , Age^2 and Age reflect the best fitting age model. $M > F$ denotes larger volumes in males compared to females. Individual male (blue) and female (pink) subjects are represented by individual lines connecting dots, and subjects measured once are represented by single dots. Adapted from Canada, Botdorf, & Riggins, 2020.

